Performance Evaluation Of Upgraded Data Centre Infrastructure

The demands on data centres have increased tremendously with the introduction of new technologies such as artificial intelligence, the Internet of Things, and now autonomous vehicles. Besides, video, cloud computing, and other bandwidth-intensive applications are forcing operators to rapidly increase network capacity. So network equipment manufacturers now need best-in-class test equipment to support them

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Emerging technologies such as the Internet of Things (IoT), 5G, artificial intelligence (AI), virtual reality (VR), and autonomous vehicles generate massive data in a network, creating new computing and performance demands on data centres. So, data centre operators, and cloud and service providers need to leverage cloud-enabled network services and innovations to deliver next-generation apps and services from anywhere. This need for virtualisation and cloud computing requires validation and scalability for reliable performance of data centre infrastructure to ensure cloud service excellence.

Data centre infrastructure is constantly evolving and needs to be optimised across all product development stages—from design, installation, to maintaining new networking products. Modernising the network architecture is required to achieve higher data rates, increase port counts, and lower cost per bit. However, this growth can be an intricate process with rapidly changing technologies.

Organisations require end-to-end testing solutions that can test, measure and assure the IP networks and deployed services performance. Now the challenge is to evaluate the deployments that will play an integral part in successful evolution and operation. Let us check some new technologies and approaches available to the data centres for network testing requirements.

Need for data centre testing

Just having all components in place will not be enough to deliver on service-level agreement (SLA) uptime assurance without regular data centre testing. The only way to know how well your in-use and redundant systems will perform is to test them regularly.

Data centre testing can help detect potential faults and vulnerabilities in a facility's infrastructure. This permits actions in advance, if any of them may contribute to server downtime or a complete system outage throughout the data centre. Regular testing of backup systems is also vital to ensure these do not fail in critical moments due to being idle or redundant for a long period of time.

Data centres should conduct regular load bank testing to put their infrastructure through the paces in addition to regular inspections, both through intelligent monitoring and visual inspections. Once a new data centre application is installed, or cloud services are moved to data centre, performance evaluation gives insight into how an application delivers under different relevant scenarios, which can help optimise deployment further.

Depending on the functional area of the data centre under test, there